

Erdős Problem #489

EP-489

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subseteq \mathbb{N}$ be a set such that $|A \cap [1, x]| = o(x^{1/2})$. Let $B = \{n \geq 1 : a \mid n \text{ for all } a \in A\}$. If $B = \{b_1 < b_2 < \dots\}$ then is it true that $\lim_{x \rightarrow \infty} \frac{1}{x} \sum_{b_i < x} (b_{i+1} - b_i)^2$ exists (and is finite)?